

Time-domain (serial) pilot: run package audit

the complete training and inference protocol, and every difference from the certified
frequency-comb $|Y|$ run

2026-08-24 (v3 — final: single-file freeze, full equations, completed 2×2)

1 Claim under test

In time-serial encoding each product traverses the mixer at its own amplitude, so the device transfer curve acts *inside* the summation and cannot be corrected at readout. The experiment compares two networks of the identical certified architecture, executed on the identical hardware protocol: one trained assuming an ideal multiplier (“clean”), one trained *through* a measurement-calibrated Shockley-like digital twin (“twin”). $2 \times 2 = \{\text{clean, twin}\} \times \{\text{digital, hardware}\}$. **No noise injection in either arm; no bias correction anywhere.** Operating point (0,0) dBm commanded, both ports; chain: the 10 dB IF pads as installed.

2 Run package inventory (single-file freeze)

As in the certified $|Y|$ freeze, **one file carries every algorithmic step**; everything else is hardware orchestration, frozen inputs, or saved outputs.

file	role	status
<code>miwen_serial_frozen_reference.py</code>	THE frozen algorithm file — preprocessing, weights loading, ideal and twin-forward digital execution, twin surface, slot encoding constants, chirp sync, slot decode, interpass, decision. Self-checks via <code>--verify</code> : weight and twin-model sha pins, clean ideal digital 99.05/99.50, bit-exact twin-surface probe hash, bit-exact hardware-capture replay decode hash	frozen; header comments list every difference from the $ Y $ freeze
<code>serial_nn_runner.py</code> (v3)	hardware session driver <i>only</i> (USRP I/O, captures, chunking, gates, incremental saves); each algorithmic step is the reference file's	frozen, hardware-validated (pack-test repeats 0.1155/0.1160/0.1162/0.1171)
<code>r35_r3plus_s0_hw.npz</code>	clean arm weights (sha 8a4c9efddb8cfd0)	verified: digital 99.50 on the battery frame
<code>serial_twin_s0_hw.npz</code>	twin arm weights (sha 825ab4ee148dc1cd), 15-epoch sprint (val-under-twin 99.87); 60-epoch sibling archived (<code>..._ep60_hw.npz</code> , val 99.92) — the 15-vs-150-epoch asymmetry vs the clean arm is disclosed	fielded for the completed N=300
<code>serial_twin_model.json</code>	the frozen twin surface (sha b07bb82b328d9c8e), produced by <code>serial_twin_fit.py</code>	held-out 0.12 dB; serial-slot cross-validation 3.7% worst bin
<code>serial_cal_{arm}_20260823.npz</code>	frozen per-arm calibration (scales + gains)	complete, both arms
<code>serial_nn_{arm}_{i0}_{i1}_20260823.npz</code>	chunk-saved predictions	complete: clean N=300 (4 chunks), twin N=300
<code>battery_random1200_idx.npy</code>	evaluation images (same frozen set as all campaign benchmarks)	frozen

3 Encoding and readout (inference protocol)

One product per 32-sample slot ($3.2\ \mu\text{s}$): non-negative activations as RF slot envelopes on subcarrier $+2.2\ \text{MHz}$; weights as LO slot envelopes on $-1.8\ \text{MHz}$, **sign and phase carried on the slot carrier** (complex weights encoded directly — both quadratures used, as in the certified $|Y|$ era). Every product lands on the single difference tone; 8-sample guards at slot edges. Chirp preamble for timing only. Readout: the slots of one output row are summed *coherently* (the digital emulation of an analog integrate-and-dump at the IF), then the **magnitude** is taken — so the absolute capture phase cancels and *no phase reference is used or needed*. Between layers: the certified $|Y|$ interpass unchanged (folded BatchNorm, ReLU, maxpool, per-image max-norm); decision $\arg \max S^2$. Calibration per arm: one frozen drive scale per layer (the layer's slot-stream RMS, fixed from a 10-image batch so every chunk runs at one operating point) and one scalar gain per layer (least squares of $|\text{row sum}|$ vs ideal). Chunks are row-aligned (no row ever spans two captures). No debias: the certified chain's Rician correction is unnecessary at row-sum SNR, and its absence is deliberate and gated.

4 The twin (training physics)

Mingran’s Shockley-derived core with the documented corrections — two distinct one-pole knees (May: LO $+2.17 \pm 0.19$, RF $+3.66 \pm 0.20$ dBm), the reciprocity-sign fix, effective (not physical) thermal voltage — plus $K=20$ tanh ridges for the map-vs-equation residual (May: 0.89 dB held-out; adding ridges beyond $K \approx 30$ buys nothing against the 0.65 dB repeat floor). Coefficients are **fitted to our port-calibrated USRP CW map**, with the datasheet and the May parameters as external plausibility gates — fitted values, not datasheet values, because family-typical specs carry 2–3 dB tolerance against a 0.9 dB model budget. Lift to a signed slot transfer by exact ring odd symmetry: $f(x, w) = \text{sgn}(xw) A(|x|, |w|)$. Excluded, per plan: thermal noise and memory. Leakage is handled exactly as in the code: the *fit*’s measurement model for CW-map cells augments the product term with two per-port leakage powers ($10^{(l_1+P_{LO})/10}$ and $10^{(l_2+P_{RF})/10}$; fitted: LO-side absent at -210 , RF-side ≈ -102 dB) — instrument terms that de-bias the map reading and are **excluded from the fielded twin**, whose slot decode rejects that content by construction. Gates as finally frozen: G2 held-out ≤ 1.3 dB on the product-dominated fit region, both ports ≥ -35 dBm (achieved: **0.12 dB**); G3 the assembled twin must *predict* the independently measured serial slot curve at $(0, 0)$, duty-corrected, fitted on none of that data (achieved: **3.7% worst bin**). G1 — named knee values vs the May fit — was **waived and demoted to reported-only**: the measured surface’s compression is joint (sum-type, cf. the May analysis’s own sum-oriented ridges), so separable per-port knees are not identifiable on this map; an honest physical finding, documented rather than forced. In training the surface is evaluated from a 256^2 bilinear table (max deviation 0.069 dB from the analytic form, verified). The twin enters **training only**: the forward applies f to every product before summation; at inference the real mixer applies the real f . Nothing twin-related exists in the inference path of either arm.

5 The equations, before and after the corrections

Instantaneous device law (draft-era form; retired)

The published manuscript’s device law (its Eq. 2) is the symmetric $\sinh \times \sinh$ ring transfer, which contains no feedthrough half-term and needs no correction; the fielded sign convention $\text{sgn}(xw)$ is that law’s odd symmetry. The form below, and the corrections applied to it, belong to a pre-arXiv draft and are retained here as provenance only ($w = \text{LO port}$, $x = \text{RF port}$, $V_T = k_B T / e \approx 25.85$ mV):

$$y(t) = \frac{w(t)}{2} + \frac{V_T}{2} \ln \left[\frac{e^{x(t)/V_T} + e^{-w(t)/V_T}}{e^{x(t)/V_T} + e^{w(t)/V_T}} \right]. \quad (1)$$

Corrections established by the May analysis and the August refinement: (i) the *effective* thermal voltage must be used, $V_T \rightarrow V_T^{\text{eff}} \approx 165$ mV ($\approx 6 \times$ the physical value; port-to-diode voltage division $\sim 1/6$ through transformers and matching); (ii) the explicit feedthrough half-term carries the wrong port — the formula’s predicted LO $\leftrightarrow$ RF asymmetry *anti*-correlates with the measured surface ($r = -0.66$); swapping the port roles in that term,

$$y(t) = \frac{x(t)}{2} + \frac{V_T^{\text{eff}}}{2} \ln \left[\frac{e^{w(t)/V_T^{\text{eff}}} + e^{-x(t)/V_T^{\text{eff}}}}{e^{w(t)/V_T^{\text{eff}}} + e^{x(t)/V_T^{\text{eff}}}} \right], \quad (2)$$

flips the asymmetry correlation to $r = +0.64$.

Power-surface hybrid (May form, AWG dataset)

$$P_{\text{IF}}(P_{\text{LO}}, P_{\text{RF}}) = \text{w2dBm}[\alpha S(P_{\text{LO}}; P_{\text{LO},\text{sat}}) S(P_{\text{RF}}; P_{\text{RF},\text{sat}}) + P_{\text{leak}}] + \sum_{i=1}^{10} w_i \tanh(a_i P_{\text{LO}} + b_i P_{\text{RF}} + c_i), \quad (3)$$

$S(x; x_{\text{sat}}) = x/(1 + x/x_{\text{sat}})$ (one-pole, linear power); fitted knees $P_{\text{LO},\text{sat}} = +2.17 \pm 0.19$, $P_{\text{RF},\text{sat}} = +3.66 \pm 0.20$ dBm, $\alpha = 27.62 \pm 0.18$ dB, $P_{\text{leak}} = -73.87 \pm 0.07$ dBm — all matching the ZEM-4300+ datasheet within 2–3 dB.

The frozen serial twin (final form, USRP map)

Product term only, fitted on the product-dominated region (both ports ≥ -35 dBm; the feedthrough-dominated edges are excluded because the slot decode rejects carrier-frequency content), $K = 20$:

$$P_{\text{prod}}^{\text{dB}}(P_{\text{LO}}, P_{\text{RF}}) = a_0 + 10 \log_{10}[\tilde{S}(P_{\text{LO}}; k_{\text{LO}}) \tilde{S}(P_{\text{RF}}; k_{\text{RF}})] + \sum_{i=1}^{20} w_i \tanh\left(\frac{a_i P_{\text{LO}}}{40} + \frac{b_i P_{\text{RF}}}{40} + c_i\right), \quad (4)$$

with $\tilde{S}(P; k) = u/(1 + u)$, $u = 10^{(P-k)/10}$. The knee values are *reported, not gated*: the measured compression is joint (sum-type; per-port knees unidentifiable on this surface). Held-out 0.12 dB on the region; serial-slot cross-validation 3.7% worst bin. Lift to the signed slot transfer by exact ring odd symmetry, with the frozen drive mapping (unit-RMS slot streams at commanded $(0, 0)$):

$$f(x, w) = \text{sgn}(xw) \cdot 10^{P_{\text{prod}}^{\text{dB}}(p_{\text{lo}}(|w|), p_{\text{rf}}(|x|))/20}, \quad p(a) = 20 \log_{10} \frac{a}{\text{rms}(\text{stream})}. \quad (5)$$

In training, f is applied to every product before summation (evaluated from a 256^2 bilinear table, max deviation 0.069 dB); at inference the mixer applies the real f ; nothing twin-related exists in either arm’s inference path.

6 Differences from the certified comb $|Y|$ run

	certified comb $ Y $ (98.83/98.92)	serial pilot
encoding	≤ 800 simultaneous tones per symbol	one tone; D consecutive time slots
where the sum happens	inside the mixer/FFT, <i>before</i> the nonlinearity’s effect is read	across time slots, <i>after</i> the nonlinearity acts per product
governing physics	Bussgang: crowd-linearized, one κ + uncorrelated deviation	per-product transfer curve; Bussgang premise absent by construction
training	noise-injected (ns25, flat 25%)	no noise ; clean vs twin-in-the-loop
calibration	43–128 per-column gains per layer + Rician debias (σ^2 estimation)	one scalar per layer; no debias
phase handling	readout magnitude per FFT bin	readout magnitude per row sum; phases still carried at TX in both
drive normalization	frame-RMS (sync-duty sensitive)	frozen per-layer slot-stream RMS (duty-free)
operating point	-35 dBm-class (record era)	$(0, 0)$ commanded
throughput	~ 2.4 h per $N=300$ arm	~ 6 h per $N=150$ half (linear in products)
evaluation images	frozen battery indices	<i>same</i> frozen battery indices
architecture, weights class	r3plus, complex weights, $ Y $ inter-pass	identical (clean arm literally the certified-era checkpoint)

7 Session discipline and gates (all pre-registered)

Pack-test at session start: 150k random products through a full-size frame; gain vs the frozen big-frame reference ($0.1155 \pm 20\%$; measured session-to-session repeat 0.6%) and residual shape in $[0.30, 0.75]$ — the decode-corruption detector. Chain gate: first-capture payload peak within band of the 2026-08-21 diagnostic. Sync margin ≥ 50 (typ. ~ 1750). Calibration saved and reloaded across the split halves (H1 = battery images 0–149, H2 = 150–299) so both halves run identical frozen corrections. No accuracy gates on either arm: both outcomes reportable. Pre-pinned digital predictions (computed before the corresponding hardware data is examined): clean digital 99.50 (verified); clean-under-twin, twin-under-twin, twin-under-ideal — pinned once the twin passes G1–G3.

8 Measured results (final)

Pinned digital predictions (committed before the corresponding hardware): clean 99.50; clean-under-twin **0.67%**; twin-under-twin **98.33** (N=300 frame) / 98.56 (full). Completed hardware, N=300, one operating point (0, 0) dBm:

	digital	hardware
clean (ideal-multiplier training)	99.50	6.33%
twin-in-training	98.33 (under twin)	97.67

The twin arm lands **2 images** from its pinned prediction ($98.33 \rightarrow 97.67$); the clean arm collapses to near the pinned clean-under-twin forecast (0.67 forecast, 6.33 measured — both “destroyed”, the residual gap consistent with hardware noise the twin deliberately omits). Per-layer calibration rel-RMSE against each arm’s own training forward: twin $[0.142, 0.087, 0.021, 0.162]$ vs clean-vs-ideal $[0.53, 0.61, 0.32, 2.94]$ — the twin describes the hardware an order of magnitude better *at every layer*. Supporting: per-product ENOB at (0, 0), naive 2.53 bits vs twin-inverted **4.38 bits** (the recoverable-vs-lost information split); Stage A drive ladder linear-residual 0.21/0.42/0.52/0.59/0.67 across -9 to $+7$ dBm — the per-product curve steepening the network must survive.

Fresh-segment replication and the second operating point (2026-08-26; byte-frozen code, frozen calibrations reloaded, digital comparators committed pre-hardware)

Every arm re-measured on the disjoint battery segment (images 300–599), plus the clean arm at a second operating point (LO -10 , RF -24) dBm (working-group geometry; runner byte-identical except the operating-point constant and re-anchored session gates, per the session pre-spec):

arm	images 0–299	images 300–599	pooled (N=600)
(0, 0) clean	6.33	5.00	5.67
(0, 0) twin	97.67	99.33	98.50
($-10, -24$) clean	81.67	82.00	81.83

The twin arm’s fresh segment (99.33) exceeds its own digital comparator (98.67) by two images — the certified-era measured-above-digital elevation, replicated on disjoint data. The collapse replicates with an identical constant-classifier fingerprint (297/300 predictions in one class both segments). The ($-10, -24$) clean result replicates across segments within one image (81.67 vs 82.00) against pipeline forecasts of 83.00/85.67 — the no-breakdown demonstration at reduced drive, with no twin anywhere in its path. Eight consecutive sessions repeated the pack-test gain within 1.6% at (0, 0) and 0.9% at ($-10, -24$).

Session trail (original, retained)

Pack-test: $|g| = 0.1162$ vs reference 0.1155, residual 0.527 — packing validated on hardware. The sync-duty drive effect (~ 3 dB payload shift between 34%-duty diagnostic frames and 0.35%-duty production frames) was caught by the pack gate, diagnosed, and re-baselined — same physics class as the historic hot-sync discoveries. Clean-arm conv1 calibration: $\text{relRMSE}(|Y|) = 0.859$ against its own ideal-multiplier model — **the serial per-product distortion, un-launderable by a scalar gain, exactly as predicted** (comb-era equivalents: 0.10–0.20). Twin fit: physics-core identifiability required the May three-stage protocol and exclusion of ADC-censored cells; iteration ongoing — current status and all intermediate failures are logged in the session record, not hidden.

9 Reading of the outcome

The two twin-validation checks both passed: (i) clean-hardware (6.33) landed at the pinned clean-under-twin forecast’s “collapse” verdict — the twin *forecast* an untrained network’s fate; (ii) twin-hardware (97.67) landed 2 images from twin-under-twin (98.33) — twin-in-the-loop training *is* the serial regime’s hardware-awareness mechanism. Together with the comb program (where the same device curve in training produced 2–10% pilots and linear calibration sufficed at 98.83), the pair of experiments demonstrates that *the encoding, not the device, decides which abstraction training needs*: nonlinearity outside the sum is a readout detail; nonlinearity inside the sum requires a physical twin in the training loop.